GridSpike: Defense-aware stress-testing of federated edge intelligence for event classification in IEC 61850 substations

Abstract

Distribution utilities need event classification on substation telemetry – for disturbance triage, targeted inspection and reliability analytics – but no utility holds enough labelled events to train one alone, and pooling measurements collides with confidentiality and critical-infrastructure rules. Federated learning is the institutional answer, provided the federation survives heterogeneous data, a bounded edge and adversarial participants. We present GridSpike, a federated framework and open benchmark – five feeders in quasi-static OpenDSS, non-IID by feeder identity and Dirichlet label skew – stress-tested. Three classifiers – convolutional–recurrent, spiking, selective state-space – learned over ten substations behind a Byzantine-admission pipeline: fidelity scoring, simulated PBFT agreement, robust aggregation, hash-chained per round. Across an expanded battery of 6,836 run records the models trace an accuracy–energy–latency trade-off. On the leakage-controlled chronological split – our headline protocol – the reference and the state-space model tie (macro-F_1 0.596 vs. 0.590, paired $p = 0.56$), and the spiking model – at 0.7% of the reference's energy – collapses to 0.13; the higher 0.77/0.60/0.16 of the random split is a leakage-inflated ceiling. Only the reference's GPU p99 (3.3 ms) fits a 4 ms fast-message envelope (82%); the others exceed it. Under a defense-aware adversary the pipeline beats neither plain averaging nor robust aggregation alone – it buys auditability, not accuracy – and usable local differential privacy is incompatible with the gate. In-cone magnitude at 10× breaks plain averaging, not the geometric median: honest rejection falls 0.80 → 0.48, attacker admission does not. The classifier provides event awareness and takes no part in trip decisions. Batteries regenerate from committed manifests; the original battery's configuration hashes verify at 67%.

Keywords: Federated learning, State-space models, Spiking neural networks, Byzantine robustness, IEC 61850, Power distribution, Differential privacy

Highlights

- Two dense model families tie on accuracy inside a three-way edge trade-off
- A quasi-static OpenDSS generator makes the federation non-IID by construction
- A high-magnitude in-cone attacker breaks plain averaging, not the geometric median
- Under a defense-aware attacker, agreement buys auditability, not model accuracy
- Usable local differential privacy is incompatible with the Byzantine admission gate

1. Introduction

Substation gateways conforming to the IEC 61850 family publish time-aligned electrical measurements – per-phase voltages and currents, power, zero-sequence quantities, frequency – that carry the signature of the events an operator cares about: ground faults, phase-to-phase faults, cold-load pickup, sustained sags and swells. Classifying those events *at the substation edge* shortens the path from measurement to situational awareness: it feeds control-room triage, inspection scheduling and reliability analytics. It is an *event-awareness* function, not a protection function – the classifier labels windows in a telemetry stream and takes no part in any trip decision. At the 60 s cadence this telemetry delivers, one T=64 window spans about an hour of acquisition, which fixes the timescale everything here operates on.

Four constraints make the target hard, and they are usually studied one at a time. *Data cannot be pooled*: no single distribution utility holds enough labelled events to train a deep classifier, and pooling across utilities collides with confidentiality and critical-infrastructure rules – the natural fit is cross-silo federated learning

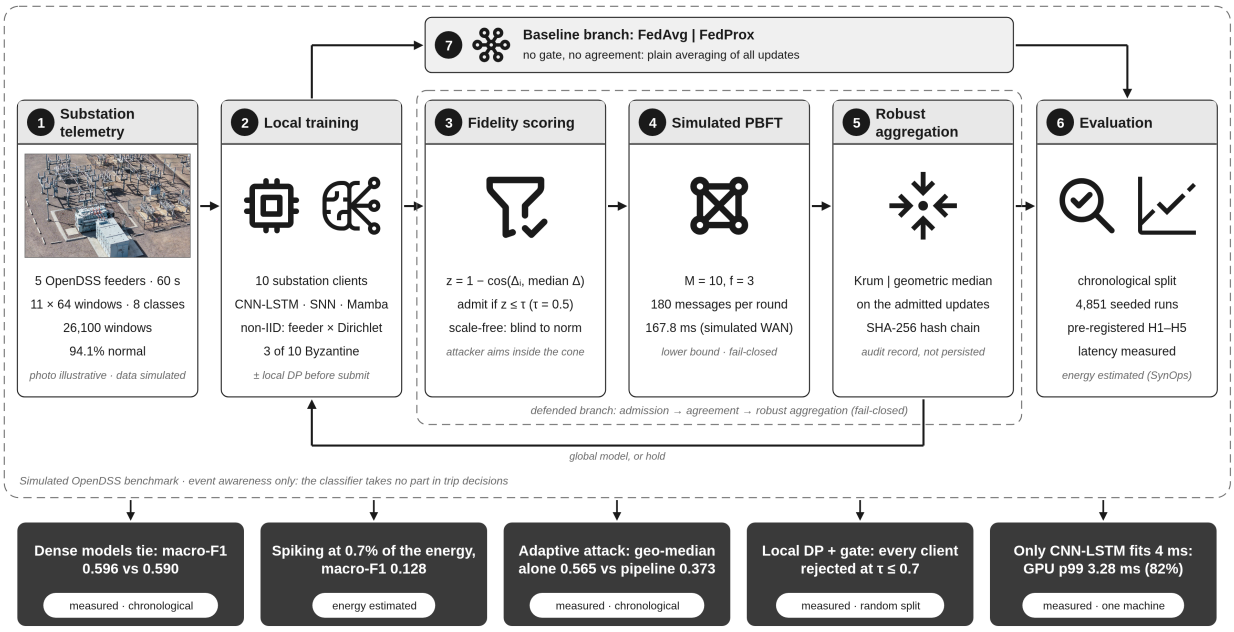


Figure 1: GridSpike overview: substation telemetry (1) through local training (2), fidelity scoring (3), simulated PBFT (4) and robust aggregation (5) to evaluation (6), against the FedAvg/FedProx baseline (7); bottom, headline results with evidence tags. Photo illustrative (CC0).

[1], a few institutional participants with substantial local data and no willingness to release it. *The federation is non-identically distributed by construction*: substations sit on different feeders with different load mixes, generation penetration and event populations, so heterogeneity is the physical situation, not a nuisance parameter. *The edge is energy- and latency-bounded*: devices are fanless and retrofit-powered, the stream is overwhelmingly normal operation – the regime an event-driven spiking classifier should exploit [2, 3] – and a classifier co-resident with protection-adjacent signalling must fit inside the 4 ms fast-message transfer-time yardstick the intrusion-detection literature already uses [4, 5]. *The federation is adversarially exposed*: a compromised gateway is a Byzantine participant, a sophisticated one knows the defense it faces [6, 7], there is no trusted server-held clean dataset [8], and the local-differential-privacy mechanism that makes participation acceptable makes honest clients look like attackers [9].

This paper presents GridSpike, a framework that handles all four constraints in one system, together with the physics-based benchmark generator that produces its data (Figure 1 draws the end-to-end path). GridSpike is a research artifact: the simulation, federated engine, attack suite, validation pipeline, experiment runner and analysis code are released together, and the batteries behind Section 5 regenerate from committed manifests, with per-run configuration hashes verifying canonically for 67% of the original battery's records.

1.1. Contributions

1. **A reproducible, physics-grounded federated benchmark generator** (Section 3). Five standard feeders solved in quasi-static time series with sample-accurate event injection, a modelled measurement chain and a realistic rare-event prior; the federation is non-IID on two axes – feeder identity and Dirichlet label skew [10]. We claim the generator, not the task [11, 12, 13].
2. **A three-architecture edge comparison under robust federation** (Sections 2, 5). A convolutional–recurrent reference, a surrogate-gradient spiking network and a selective state-space (Mamba) model on the same benchmark and admission pipeline – a federated spiking classifier on substation measurements, and a state-space model in this setting, have no precedent [14, 15, 16]. The honest finding is a trade-off on the leakage-controlled split: the two dense models tie, the spiking model is two orders of magnitude cheaper in energy but collapses, and only the reference fits a co-resident 4 ms envelope.
3. **A defense-aware attacker carried end-to-end** (Section 2.5). An adversary that knows the admission rule constructs in closed form an update inside the admissible cone, plus a separate magnitude lever exploiting the scale-freeness of cosine scoring [6, 7, 17, 18], evaluated against both robust

aggregators with the evasion-versus-damage asymmetry measured.

4. **Consensus overhead measured next to the accuracy it buys** (Sections 2, 5). Exact message counts and simulated wide-area latency per round, plus the ablation the agreement-plus-aggregation literature [19, 20] usually omits: the pipeline against the robust aggregator standing alone. Agreement buys auditability, not accuracy.
5. **An empirical privacy-versus-admission account** (Section 5). That differential privacy composes poorly with Byzantine resilience is a theorem, and not ours [9]; we provide the measured account of honest false-rejection against attacker admission across noise multiplier and threshold, annotated with the Rényi privacy budget [21]. At every usable privacy level the gate fails closed: the mechanisms are incompatible, not tunable.
6. **A compute-share budget for waveform-window classifiers** (Section 5). Measured latency percentiles for all three models against the 4 ms fast-message envelope – the adjacent literature benchmarks packet-level detectors against that figure [4, 5]; waveform classifiers have not been placed in it before.

1.2. What this paper does not claim

Stating the boundary is cheaper than being corrected at it. We do not claim first federated learning for power-system fault classification [11, 12], the discovery of the privacy–robustness tension [9], or novelty in combining Byzantine agreement with robust aggregation [19, 20]. We do not claim a hardware energy measurement (figures are SynOps estimates [3], arithmetic only), field validation (data come from a simulator; Section 7), or end-to-end verification of the hash-chained ledger (recomputed by property tests, never persisted). We do not claim Byzantine-fault-tolerant security under equivocation or partition (the agreement stage is simulated), a protection function (the 4 ms figure is a compute-share reference, not a deadline), or that the spiking classifier beats the reference – the design target is accuracy *parity* at lower estimated energy, tested as an equivalence claim.

1.3. Related work

Five active lines meet here; each is developed fully, with its most dangerous neighbour named, in Supplementary Section S2 and Table S1. *Federated grid analytics* is established [11, 12, 13] – but every member consumes a fixed dataset; none constructs the federation from physics. *Spiking classifiers for power systems* exist on phasor measurements [16] – centralized, the strongest precedent for our energy claim – and federated spiking learning exists on vision benchmarks [14, 22], recently under Byzantine attack [15] but with a non-omniscient adversary and no consensus layer. *Byzantine-robust aggregation* is settled [23, 24]; trust bootstrapping [8] is excluded (no clean server-side data exists); defense-aware poisoning defines our adversary [6, 7, 17, 18]. *Agreement-plus-aggregation* combinations are crowded [19, 20] but report neither the protocol's overhead nor a defense-aware attacker. *DP–Byzantine composition* is theoretically understood [9] and actively engineered [25], but no member of that line supplies a measured operating curve in an application system. *Latency against the 4 ms envelope* is standard only for packet-level intrusion detection [4, 5]. The claim is not any column of that space but the intersection: no prior system occupies it.

Formal definitions, the extended related work, the realism parameter audit, battery and provenance detail, and the complete secondary-results battery are in the supplementary material.

2. System architecture and threat model

GridSpike is organized in four layers (Figure 1 draws L1 as stage 1, L2 as stage 2, L3 as stages 3–5 and L4 as stage 6); what follows describes what the released implementation does. The formal machinery it summarizes – federated objective and FedProx term, LIF neuron and surrogate, input codings, Krum and geometric-median aggregators, PBFT quorum and message accounting, Gaussian/Rényi local DP – is collected with its derivations in Supplementary Section S1.

2.1. L1 – physics and data generation

Layer 1 turns feeder models into labelled per-substation windows (Section 3). The sample contract is fixed for every model, defense and experiment: $x \in \mathbb{R}^{11\times 64}$ with channels $(v_a, v_b, v_c, i_a, i_b, i_c, p, q, v_0, i_0, f_{\text{dev}})$ and $y \in \{0, \ldots, 7\}$ over NORMAL, FAULT_LG, FAULT_LL, FAULT_LLG, FAULT_3PH, OVERLOAD, SAG, SWELL.

2.2. L2 – local training, privacy and poisoning

The federation runs *in process*: all K trainers, the attack and privacy hooks, the pipeline and the evaluation execute sequentially under one seeded random-number tree – a distributed runtime (Flower [26], the declined alternative) cannot guarantee bit-level determinism of an attack-and-consensus pipeline under seeded randomness, the same reason the Byzantine-robust literature

Algorithm 1 One GridSpike validation round

Require: $\{u_i\}_{i=1}^{n}$ (full parameters); $w^{(t)}$; threshold τ; validators M, faults f; aggregator A

$\Delta_i \leftarrow \text{flat}(u_i) - \text{flat}(w^{(t)})$ ▷ direction, not model
$\bar{\Delta} \leftarrow \text{median}_{\text{coord}}(\Delta_1..\Delta_n)$
$z_i \leftarrow 1 - \cos(\Delta_i, \bar{\Delta})$ ▷ scale-free ballot, $\in [0, 2]$
$P \leftarrow \{i : z_i \le \tau\}$ ▷ primary's proposal
$(c, A^*, n_{\text{msg}}, \hat{\imath}) \leftarrow \textsc{Pbft}(M, f, z, \tau)$ ▷ Eq. S10
if $\neg c$ **or** $A^* = \emptyset$ **then**
  ledger.append⟨round, z, NO-COMMIT⟩
  return $w^{(t)}$ ▷ fail-closed: hold the model
end if
$w^{(t+1)} \leftarrow A(\{u_i\}_{i \in A^*}, f)$ ▷ robust aggregate
ledger.append⟨round, z, A^*, n_{msg}, $\hat{\imath}$⟩ ▷ SHA-256 chain
return $w^{(t+1)}$ ▷ engine strategy bypassed

runs custom in-process loops [6, 8, 7]. The cost is no real network transport; the benefit is exact reproducibility of a reported run.

Three model families run on identical footing: a **convolutional–recurrent reference** (CNN-LSTM); a **spiking classifier** (rate or temporal-contrast coding into LIF layers unrolled over T steps, trained through an arctangent surrogate, Eqs. S3–S4, with spike counters kept outside the autograd graph so Eq. S5 is auditable); and a **selective state-space model** (Mamba [27], a dense linear-time sequence model added as a strong non-spiking comparison). Under the rare-event prior all three train with class-balanced cross-entropy [28] computed from each client's *own* labels only – otherwise the unweighted minimum is "always predict NORMAL".

The per-client submission order is itself a threat-model statement: local train → local DP (clip, noise) → poison (if Byzantine) → submit. Nothing clips or noises an update after the poisoning hook, so local DP is never a bound on malicious updates.

2.3. L3 – admission, agreement, aggregation and audit

Layer 3 is a *ValidatorFn* plugged into the federated engine (Algorithm 1).

Fidelity scoring. The ballot is $z_i = 1 - \cos(\Delta_i, \bar{\Delta})$ against the *coordinate-wise* median of the round's update directions $\Delta_i = w_i - w^{(t)}$ – the cheapest location estimate with breakdown point $1/2$, needing no clean server-side reference dataset a multi-utility federation cannot have [8]. Cosine distance is scale-free, so one threshold τ survives learning rate, model size and round index; scoring the *direction* rather than the full vector is what separates an exactly reversed update from an honest one. Degenerate cases score the neutral 1.0.

Agreement. Validators run normal-case PBFT [29] over the public score vector: a committing round costs exactly $2M(M-1)$ messages (Eq. S10), 180 at the default M=10, and tolerates $f = \lfloor (M-1)/3 \rfloor$ faults. The realized fault model – faulty validators vote *against* a deterministic ballot rather than equivocating, the primary always honest, authentication and view change unmodelled – is stated exactly in Section S1.4 and makes every reported consensus cost a **lower bound**.

Aggregation and audit. The admitted set is aggregated by Krum, multi-Krum or the geometric median (Eqs. S7–S8) and the engine *bypasses* its own averaging strategy – re-averaging a robust aggregate would undo it; multi-Krum averaging is deliberately unweighted, since n_k is self-reported. Every round appends a record to a SHA-256 hash chain – tamper-evident to edits and interior deletions but not to a writer who re-hashes every successor, nor to tail truncation; and it is *not persisted*: auditability is a property of the design verified by property tests, not a measured result.

2.4. L4 – orchestration and provenance

Layer 4 exists so the numbers regenerate: a manifest enumerates the battery as fully specified configurations with SHA-256 hashes and deterministic run identifiers, and every run serializes a `RunResult` with configuration, seed, events, metrics and an environment fingerprint (audit in Section S4.6).

2.5. Threat model

$K = 10$ substations participate fully every round; up to f_{att} are Byzantine – the headline uses $f_{\text{att}} = 3$, exactly the tolerance of the $M = 10$ committee. Byzantine clients submit arbitrary vectors and may collude. Four attacks are implemented: *sign-flip* submits $w^{(t)} - s\,\Delta_k$, the loud attack every defense catches and the baseline evasion is measured against; *Gaussian noise* perturbs every parameter at local-DP-like scales – to a direction-only gate, a noised honest client and a noise attacker are the same vector, the whole privacy tension; *label flip*, data-level poisoning realized as *dynamic per-batch noise* (labels re-drawn each round at frac = 1.0 – a weaker adversary than static poisoning); and *adaptive cosine*, the defense-aware attacker.

The defense-aware attacker. Its knowledge is granted generously and stated exactly. It **knows**: the admission rule and a target score ς; the round-start global parameters; its own honest training result; and the previous aggregate, so $w^{(t)} - w^{(t-1)}$ gives the best *public* estimate of the honest direction. It **does not know**: other

clients' updates, the realized median, which validators are faulty, or any aggregator state – defense-aware but not omniscient [6, 7]. Given a unit estimate u of the honest direction and its own honest direction h, it works in angles, the cosine constraint being a cone half-angle:

$$\theta_{\text{budget}} = \arccos(1-\varsigma), \qquad \theta_{\text{err}} = \angle(h,u)/\sqrt{K},$$
$$\theta = \max\Big(0,\ \theta_{\text{budget}} - \theta_{\text{err}}\Big)(1-\eta),$$
$$\Delta^{\text{adv}} = s\,\|h\|(\cos\theta\ u + \sin\theta\ \omega), \tag{1}$$

with η a safety margin and ω a unit vector orthogonal to u, drawn identically by all colluding clients per (seed, round) – the coalition lands on a single point of the cone, and f mutually identical malicious vectors are their own dense neighbourhood, which is what defeats a nearest-neighbour selector [23]. Since $u \perp \omega$, the construction holds its score identically in one evaluation, with no query to the defense. Two precisions matter: ς is a *fixed attacker-side parameter* – the attacker never reads τ, so at $\tau = 0.5$ it aims at a fifth of the tolerated score, understating its measured evasion; and in round 0, with no previous aggregate to aim at, it **submits its honest direction**.

The magnitude lever s is deliberately separate: cosine is scale-free, so $s > 1$ submits the same direction with several times the norm, dragging an *averaging* aggregator proportionally further while the score does not move. Its default is $s = 1$ so no headline result rests on a trivially clippable norm; the lever is swept in Section 5.8. The geometry is not new [17, 18, 7]; tying it to the deployed statistic and carrying it through agreement into measured damage is.

3. Benchmark: physics, realism and federation construction

No public dataset provides per-substation labelled fault and disturbance waveforms suitable for a K-client federation; that gap is why the benchmark generator exists. Rather than assert realism, every generator parameter carries a verifiable source – a normative standard, a public dataset, or a peer-reviewed paper – and is labelled by the *kind* of evidence behind it (*measured-data*, *standard*, *fitted-synthetic*, *modeling-choice*); the complete audit, parameter by parameter, is Tables S2–S5. A test asserts that no row ships without a source, and Section 7 prices what the fitted-synthetic rows cost.

3.1. Feeders, solver and events

Five unmodified distribution test feeders are vendored – the IEEE 13-, 34-, 37- and 123-bus feeders and the IEEE PES European Low Voltage Test Feeder [30] – and clients are assigned feeders cyclically, so feeder identity is a feature-skew axis before any label skew. The IEEE 37-bus feeder is an ungrounded delta system, retained deliberately: on it a ground fault appears as neutral displacement, not fault current, so *the same label means a different signal shape* at different clients. The solver is the DSS-Extensions OpenDSS engine in quasi-static daily mode at a **60 s** step, metering at the substation bay (feeder-head VT, first-section CT); a step that fails the power flow is re-solved and then *holds the last valid value* – the repeat-last convention of telemetry front ends – at an aggregate held rate of 0.0059% (Table S7).

Residential demand uses the CREST measured-model household profiles [31]; rooftop PV sits on 40% of load buses at 15%/40% penetration with Markov cloud transients fitted to measured statistics [32, 33, 34]. Events arrive at a Poisson rate per client-day – line-to-ground, line-to-line, three-phase faults at documented resistance distributions [35, 36], sags and swells targeted to land in the IEEE 1159 bands [37, 38], and sustained cold-load-pickup overload [39] – with episode durations of 2–10 minutes.

No protection equipment is modelled. The generator contains no relay, recloser, fuse or breaker: a fault stays energized for its whole drawn duration where a real feeder would clear it in cycles. Fault windows are therefore *more* observable here than in the field, absolute fault-class accuracies are optimistic, and the benchmark supports the relative comparison across models and defenses, not a field-accuracy claim.

The class prior is realistic, and that is the hard part. The generator is tuned so $\geq 90\%$ of windows are NORMAL; the realized share is 93.0–94.1% across the battery federations (Table S6). All primary metrics are computed under this prior – macro-F_1 and per-class recall, never accuracy alone – and the rarest class can have single-digit support in a client split, so its recall is unstable and reported, not smoothed.

3.2. Measurement chain and federation construction

The classifier sees instrumented telemetry, not clean solver output: class-0.5 CT/VT error models [40, 41], a 16-bit ADC, dropout episodes at measured rates [42], and 60 s RMS gateway telemetry – not IEC 61850-9-2 sampled values, which the generator does not model. System frequency is not a solver output, so f_{dev} is a

stationary Ornstein–Uhlenbeck process fitted to measured statistics [43] with event-coupled dips – a fitted-synthetic channel whose possible label-shortcut is audited in Section 5.1.

Windows are T=64 samples at stride 16 (75% overlap), 87 per simulated day, 2 610 per client at the 30-day default. Heterogeneity is two-axis: feeder/load-shape identity gives feature skew and a Dirichlet(α) draw over the eight classes gives label skew [10] (α = 0.5 reference, $\{0.1, 1.0\}$ and a near-uniform control swept). Splits are per-client 70/15/15, seeded, the normalizer fitted on the training split only. Two protocols exist and both are reported: the pre-registered seeded *random* permutation – under which the committed `experiments/measure_leakage.py` finds 99.9% of test windows overlapping a training window – and the leakage-controlled *chronological* split (contiguous blocks, four-window purge, $\approx$ 0% overlap), adopted as the headline protocol. The consensus layer's latency is a sourced wide-area lognormal (median 25 ms) [44]. The same pipeline runs unmodified on the NREL SMART-DS systems [45], the declared scale-up path.

4. Experimental design

The design was fixed and committed *before* any result run started: a machine-readable battery manifest enumerates every planned run with its full configuration and a SHA-256 configuration hash; a pre-registration file lists the hypotheses, their directions, their tests and their decision rules; and an analysis module implements every test named below.

4.1. Reference frame and battery

Federated results are only interpretable between two bounds, and the surveyed application literature usually reports neither. We run both: a *centralized ceiling* (pooled data) and a *local-only floor* (each substation alone; mean over ten clients), beside FedAvg [1], FedProx [46] and the full pipeline with each robust aggregator – six arms × three models. The attack grid adds four defense variants – no-defense, Krum-alone, geomed-alone and fidelity-only – so the pipeline is ablated against the robust aggregator standing alone (Table S8). One family is excluded with a stated reason: root-dataset trust bootstrapping [8] assumes a clean server-held dataset, unavailable by design here.

The battery is organized in eleven blocks (Table S9): Tier 1 carries the clean reference frame (270 runs) and the attack × defense grid (900 runs, $f_{\text{att}} = 3/10$); Tier 2 carries the sweeps; Tier 3 is the *control battery*, specified on 2026-07-31 after the registered analysis was complete – the entire Tier-1 grid re-run under the chronological split (1,170 runs) and the attacker-magnitude sweep (432 runs). The merged corpus holds 4,851 distinct seeded runs from 6,836 files after two-pass deduplication, reconciled in Section S4.2.

Tier 1 and the temporal controls run **15** seeds; every Tier-2 sweep and the magnitude control run **9** – at fifteen paired seeds an exact one-sided Wilcoxon reaches 2^{-15}, so every pre-registered family is decidable after Holm correction, and one seed drives the entire stochastic pipeline, making repetitions independent dataset redraws. The X3 extension mixed two run environments (RTX 3060/3090/4080, torch 2.4–2.13 for the registered seeds; RTX 5090/4090, torch 2.9.1 for the expansion), so temporal cells mix environments by construction; a homogeneous-subset analysis on seeds 5–14 leaves the headline conclusions unchanged (Section S4.3). Attack settings are fixed throughout: $\varsigma = 0.1$, $\eta = 0.15$, $s = 1$ (swept only in Section 5.8), label-flip rate 1.0, all colluding clients sharing one ω per (seed, round) (Section S4.4).

4.2. Models, hyperparameters and metrics

All three models train with an identical budget: 30 federated rounds, one local epoch, batch 64, Adam at 10^{-3}, class-balanced cross-entropy [28]. The CNN-LSTM has convolution channels $(32, 64)$, LSTM hidden 128, dropout 0.3; the spiking model has width 256, two LIF layers, $T = 8$ unroll steps, temporal-contrast coding at 0.2; the state-space model is four selective-SSM blocks of width 96 over a stride-4 patch stem (278,216 parameters, 4,590,336 MACs per inference). This single optimizer configuration is a *budget decision*: only the spiking delta-coding threshold was swept on validation macro-F_1; every other hyperparameter was set once on clean FedAvg and frozen for every attack, defense and sweep – no arm gets a tuning advantage, and none a rescue, so the spiking collapse of Section 5 is a collapse *under this optimization budget* (plain SGD barely moved off initialization, and thresholds below 0.1 fire on the quantized measurement noise).

Every run records accuracy, macro-F_1 (headline, over classes present in the labels), weighted and per-class F_1 and the confusion matrix; under attack, degradation, attacker admission and honest false-rejection; consensus messages, simulated latency and the rejected-set trace; the Rényi budget ε (Eq. S11); SynOps, MACs, spike rate and estimated energy (Eqs. S5–S6); measured latency percentiles; and cost measures.

4.3. Statistical protocol and pre-registration

The protocol is non-parametric throughout [47], $\alpha = 0.05$, two-sided unless a direction is pre-registered: Wilcoxon signed-rank on per-seed paired differences with Holm correction [48] within each hypothesis family; Friedman with a Nemenyi critical-difference diagram for multi-arm comparisons; Cliff's δ [49] beside every Δmacro-F_1; BCa bootstrap intervals (10 000 resamples) on every mean; McNemar's exact test plus a two-one-sided-tests equivalence check at a ± 0.02 margin for matched accuracy, so parity can be a positive finding; and Jonckheere–Terpstra ordered trends with Spearman secondary.

Five hypotheses were registered before the first Tier-1 run: **H1**, the spiking classifier matches the reference's accuracy (equivalence) while its estimated energy is far lower; **H2**, under each attack the full pipeline beats no defense; **H3**, the defense-aware attacker evades the gate where sign-flip is caught (H3a) and, being admitted, costs at least as much accuracy (H3b); **H4**, honest false-rejection increases monotonically with the DP noise multiplier; **H5**, accuracy degrades with heterogeneity, and plain averaging degrades more than the pipeline. Four dated amendments are on record – an analysis-code fix restoring Mamba to the test loops, the label-flip realization statement, the post-registration control battery, and the X3 seed extension – none changing a hypothesis, metric, test or decision rule (Section S4.5).

4.4. Reproducibility and AI-methodology disclosure

Every run serializes configuration, seed, per-round events, metrics and an environment fingerprint; per-run git provenance is mostly absent (96.9% of block-labelled records), so regeneration rests on the committed manifests, configuration hashes – re-verifying canonically for 66.7% of the original battery's records – and seeded determinism (full audit in Section S4.6). Bit-reproducibility is claimed on CPU only; all latency benchmarks ran on one fixed reference machine; no model was quantized (INT8 rows are analytic re-costings). Per journal policy, AI methodology use is disclosed here: AI coding assistance was used in implementing the software artifact under author review; no AI system generated, imputed, labelled or selected any experimental data, chose hypotheses, tests or decision rules, or produced any number reported in Section 5.

5. Results

All values are means over seeds with the seed count stated, with BCa 95% intervals; per-seed raw results

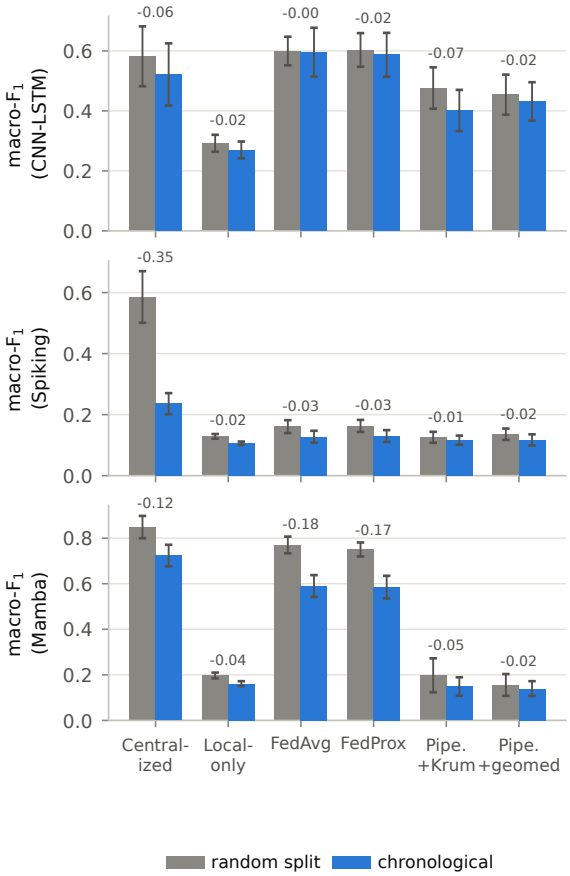


Figure 2: Reference-frame macro-F_1 (mean ± sd) per arm and model, chronological split (T1a_temporal) beside random split (T1a), 15 seeds; numbers give the unpaired change. Cells: Table S23.

ship with the artifact. macro-F_1 on the pooled global test set under the realistic class prior is the headline metric. The section leads with the reference frame under the leakage-controlled chronological split – the headline protocol – then reads the pre-registered random split as a leakage-inflated diagnostic ceiling, then reports the pre-registered hypotheses, several of which did not survive their own decision rules.

5.1. Reference frame under the leakage-controlled split

The two protocols bracket every absolute number, and differ in whether a test window may share samples with a training one: windows are cut at stride 16 over length 64 (75% overlap), and the committed measurement script finds 99.9% of test windows overlapping a training window under the random split against $\approx 0\%$ under the chronological one. Figure 2 puts both protocols side by side for the full frame (exact cells: Table S23).

The margin was leakage: the two dense models tie. The ranking survives as a tie at the top, not a lead. Under FedAvg the CNN-LSTM and the state-space model land on each other – macro-F_1 0.596 against 0.590, a paired difference an exact two-sided Wilcoxon cannot separate ($p = 0.56$) – and the spiking model sits at 0.128; FedProx repeats the shape (0.587/0.585/0.130). Most arms lose 5–25% of their random-split score and the floor moves with everything else; the spiking ceiling loses more (below). The exception is the federated CNN-LSTM, statistically indifferent to the split (FedAvg −0.6%, $p = 0.89$; FedProx −2.7%): the random protocol was crediting leakage accuracy to the state-space and spiking arms and to both bounds, not to the reference. What changes is the claim – the state-space model's earlier lead was a margin on leaky cells.

The spiking ceiling was mostly leakage. The largest single change is not in a federated row: the centralized spiking ceiling falls 0.586 → 0.236 (−60%) against −10%/−15% for the dense ceilings, on matched recipes. The model is on the floor in *both* regimes (0.236 centralized, 0.128 federated, 0.121 trivial predictor) – not a structural limit of *federated* surrogate-gradient training but the blunter finding that this architecture does not learn this task at this budget. The ceiling bounds only two of three arms, too: pooling the CNN-LSTM does not beat its federated arm (0.522 vs. 0.596 chronological, n.s.), the per-client effective-number reweighting [28] being miscalibrated on the pooled corpus.

A shallow probe, and the channel it audits. A deep model is only justified if a cheap one cannot do the job. A gradient-boosted tree at library defaults reaches macro-F_1 0.692 pooled and 0.533 under the chronological cut – at or above the CNN-LSTM's *centralized* ceiling under either protocol, so the reference is justified by timing (Section 5.4), not centralized accuracy; only its *federated* arm (0.596) edges the pooled probe (0.692 leans on default early stopping – disabled, 0.275; detail in Section S5.2). The same probe audits the sharpest benchmark shortcut: dropping the synthetic f_{dev} channel costs −0.011/−0.027 macro-F_1 (random/temporal, intervals spanning zero) while the channel alone reaches 0.295/0.234 – real but far-from-sufficient signal (Table S11). Convergence is in Figure S3: only the CNN-LSTM averaging arms reach their ceiling, on every seed.

5.2. The random split as a leakage-inflated diagnostic

What the random split was crediting is visible cell by cell in Figure 3: cells carrying real accuracy give up as much as 0.19; the pipeline cells barely move, because the random split only credits accuracy to arms that had accuracy to inflate. The two results the defense analysis rests on both reproduce under the controlled protocol: geomed-alone still recovers the loud attacks (Gaussian noise on the CNN-LSTM: 0.059 → 0.541), and the full pipeline is still catastrophic against the defense-aware attacker (pipeline+Krum holds Mamba at 0.051 where no defense holds 0.553). One confound comes with the cut: the test slice moves to a year-end segment, so every Δ mixes leakage removal with seasonal shift – the *differential* pattern argues the leakage term dominates. Both protocols therefore stay: the random-split battery (Tables S10 and S12) carries the sweeps and the preregistered verdicts, and the temporal columns are the ones to quote for absolute accuracy.

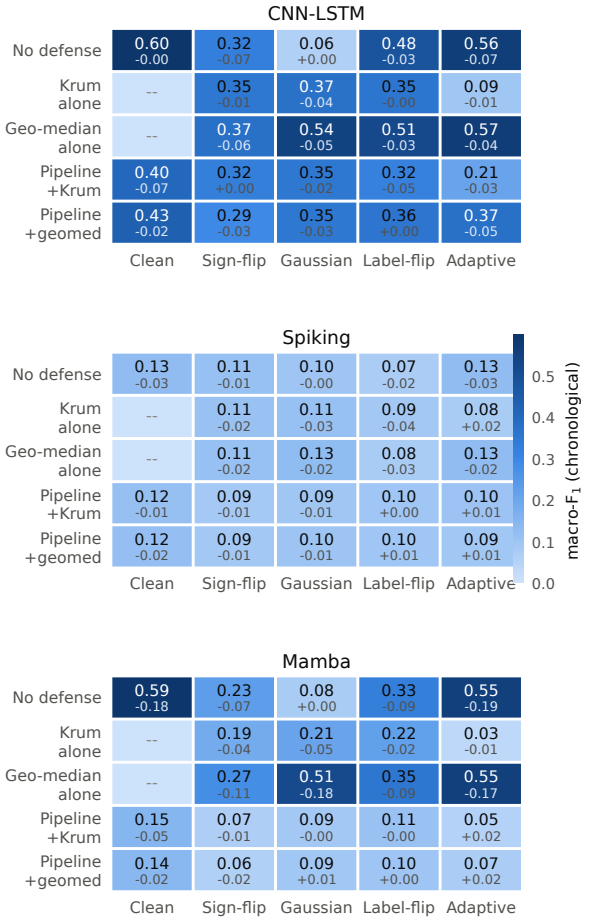
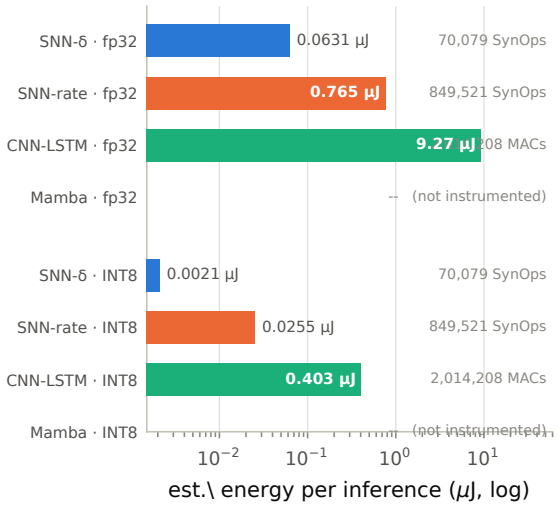


Figure 3: Attack × defense macro-F_1, chronological split (T1b_temporal, $f_{\text{att}} = 3/10$, 15 seeds), one panel per model; second line: change against the random split. Cells: Table S24.



not measured: Mamba INT8 (no MAC counter); Mamba fp32 (no MAC counter)

Figure 4: Estimated energy per inference (log scale) per model and coding at T=8, 45 nm constants [3]. Arithmetic-only SynOps estimate, not a hardware measurement. Cells: Table S25.

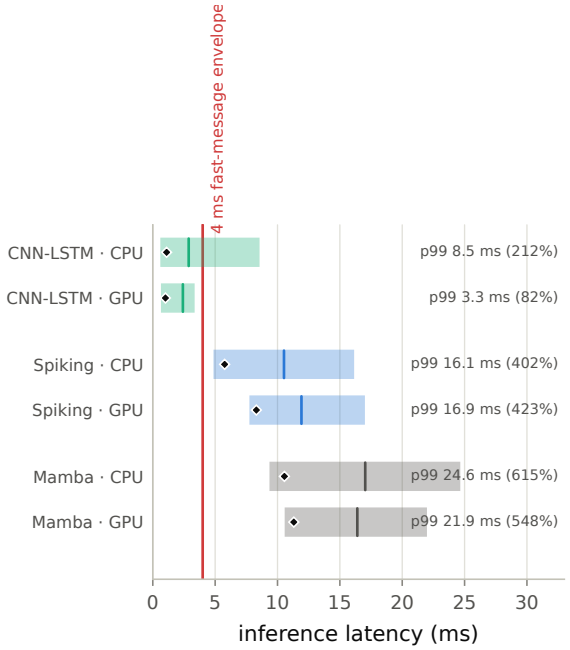


Figure 5: Measured inference latency (fp32, batch 1, n = 1000) per model and device against the 4 ms fast-message envelope; bar p50–p99, tick p95, diamond mean. A compute share, not compliance. Cells: Table S26.

5.3. *H1: accuracy parity and the energy ratio*

The pre-registered parity claim fails decisively: McNemar's exact test rejects equality ($p < 0.001$) and the paired Δmacro-F_1 lies far outside the ± 0.02 equivalence margin (BCa 95% $[-0.465, -0.417]$). What survives is the energy accounting (Figure 4): under delta coding the spiking model performs 70 079 synaptic operations per inference against 2 014 208 multiply–accumulates – an estimated 0.063 against 9.27 microjoules, a ratio $\rho \approx 6.8 \times 10^{-3}$ with the one-sided BCa interval wholly below one: H1b is supported, H1a is not, and H1 fails as a conjunction. Rate coding costs an order of magnitude more for less accuracy (Section 5.13); Mamba reads 21.1 μJ, 2.3× the reference. These are arithmetic-only estimates, not hardware measurements. The spiking model delivers its energy reduction only by conceding almost all of its federated accuracy – Figure S4 places every costed arm on the accuracy–energy plane.

5.4. *Latency against the fast-message envelope*

Inference is one term of the end-to-end chain, so a measured p99 is a *compute share* of a co-resident 4 ms fast-message envelope, not a compliance verdict on it (Figure 5). At fp32 only the CNN-LSTM leaves any of the envelope, and only on GPU: 3.28 ms p99, 82% of it, against 212% on CPU; the spiking model (402–423%) and Mamba (548–615%) exceed the whole envelope alone, and no model benefits from dynamic INT8 quantization – torch's kernels reject the recurrent and state-space packed parameters, so every measured latency is fp32 (distributions in Supplementary Figure S6). The three architectures resolve into a clean trade-off: Mamba ties for the accuracy lead but is slowest, the spiking model is cheapest but neither accurate nor fast, and the reference is the only one whose compute would fit beside a protection-adjacent function – on one fixed reference machine, which orders the models but certifies no substation device.

5.5. *H2: the pipeline under attack*

The grid does not reward the pipeline; this is the paper's central negative result (Table S12, decomposed rung by rung in Table S13). Under the defense-aware attacker, geometric-median aggregation *alone* preserves accuracy almost perfectly (CNN-LSTM −0.020 from no defense; Mamba −0.013) while rejecting nobody, whereas the same aggregator behind the gate and PBFT drops to 0.420/0.051 and rejects 66–97% of honest updates. Krum alone is worse than useless – the single-update selector is exactly what the cosine-aligned attacker games (0.043–0.100) – and the pipeline compounds the damage: Krum's guarantee needs $n \geq 2f + 3 = 9$ admitted updates at $f = 3$, and a gate rejecting most of the honest population routinely hands it fewer. Under the loud attacks a robust aggregator re-

covers most of the loss, the gate never adds accuracy over the bare aggregator, and Mamba under the adaptive attack with *no* defense holds 0.739: it is robust by itself, and the pipeline takes that away.

The Friedman test agrees under both blockings (Figure S5): at the pre-registered blocking exactly two pairs separate – geometric-median-alone above each pipeline arm – and no-defense against pipeline+Krum does *not*. Two of the 24 per-cell tests support the pipeline after Holm correction (both aggregators against Gaussian noise on the CNN-LSTM, $\delta = +1.00$) – loud attacks on the weaker model, not the defense-aware attacker the pipeline was built for (Table S21). H2 is recorded as not supported (Table S22), with the qualification that the agreement layer contributes the audit record and the measured overhead, not accuracy; the attack-fraction sweep (T2b) moves the same way.

5.6. *Opening the gate: the admission-threshold sweep*

The pre-registered $\tau = 0.5$ is one point on a curve, and sweeping it changes the diagnosis (Table S15). As τ opens to 1.0 the gate stops rejecting honest clients (FRR 0.68–0.97 $\rightarrow \leq 0.016$) while the defense-aware attacker is admitted essentially always (≥ 0.92 at $\tau \geq 0.8$) – the worst of both worlds on the registered decision rule. What it produces is not: behind the geometric median accuracy *recovers* as the gate opens (0.184 $\rightarrow$ 0.487 pooled; Mamba 0.051 $\rightarrow$ 0.718, within 0.01 of geo-median-alone), while behind Krum it collapses further (0.120 $\rightarrow$ 0.069). This revises H2 rather than reversing it: at $\tau = 0.5$ the gate was *throttling its own aggregator* – the negative result is about calibration, not mechanism, and no setting of τ makes the pipeline *add* accuracy over the bare aggregator.

5.7. *H3: the defense-aware attacker's evasion–damage asymmetry*

The mechanism behaves exactly as designed, and at fifteen seeds it gets its statistical answer (Table S14). Against either aggregator the defense-aware attacker holds its realized score to 0.40–0.55, straddling $\tau = 0.5$, where the loud sign-flip scores 0.98–1.05 and is rejected outright: admission rises from 0.000 to 0.53–0.59 and to 0.87 against Mamba, at $p_{\text{Holm}} = 1.8 \times 10^{-4}$ in all six cells – the evasion is real and H3a is supported. The same gate still rejects 66–98% of *honest* updates: it does not separate friend from foe, it rejects both. The damage leg splits by aggregator: behind Krum the admitted attack costs *more* than the rejected sign-flip on two of three models (0.242 vs. 0.153; 0.167 vs. 0.122, both significant); behind the geometric median it runs backwards

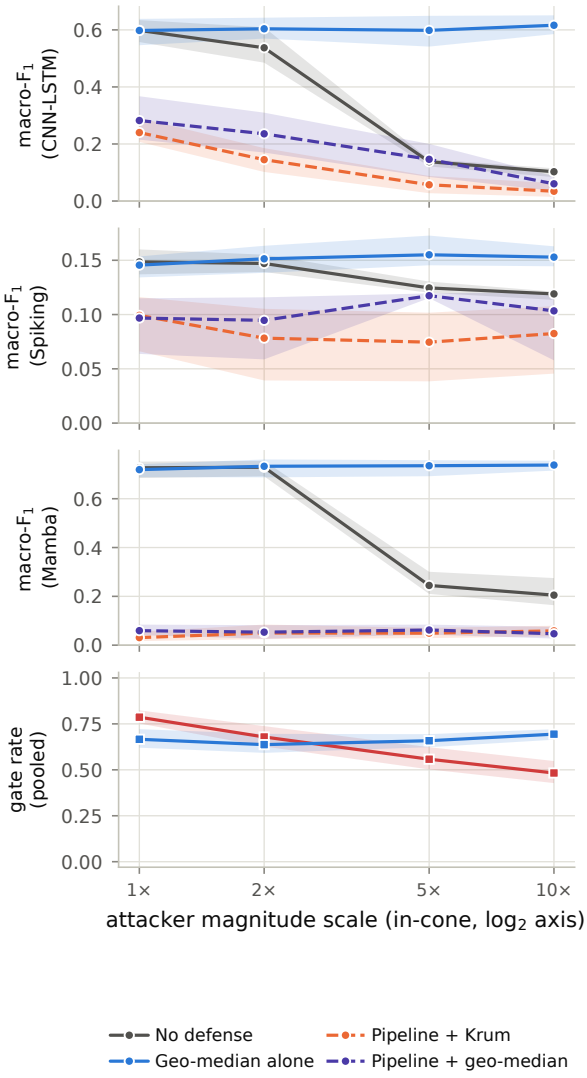


Figure 6: macro-F_1 (BCa bands) and pooled gate rates as the in-cone attacker scales its update by $s \in \{1, 2, 5, 10\}$ (T4mag, random split, $f_{\text{att}} = 3/10$, 9 seeds); s=1 is the default. Cells: Table S27.

(0.034 vs. 0.134), because a robust average is not moved by an update it was content to admit. H3b stays not supported on its 2/6 cell count – a property of the averaging aggregator, not of the test's resolution.

5.8. *The attacker's magnitude lever*

Cosine scoring is scale-free, so an in-cone attacker can multiply its update without moving its fidelity score; the pre-registered battery left that factor at 1 throughout, and block T4mag sweeps it (Figure 6). Three things happen, and they point the same way. *Plain averaging breaks*: the undefended CNN-LSTM falls 0.598 $\rightarrow$ 0.103 and Mamba 0.727 $\rightarrow$ 0.205 – significant at 5×–10× on the dense models, descriptively the damage leg

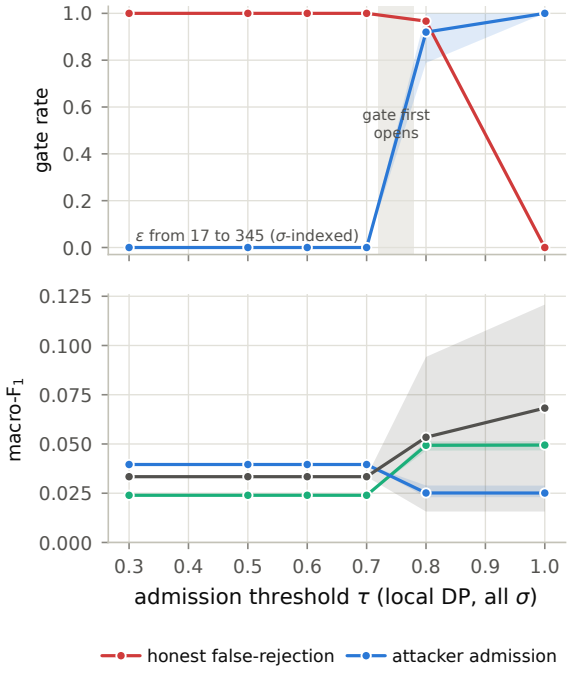


Figure 7: Gate rates (top) and macro-F_1 (bottom) against τ on pipeline+Krum under the defense-aware attacker, pooled over four σ (band: cross-σ range; T2c and T3τ, random split, 9 seeds). Cells: Table S28.

H3b failed to find at unit magnitude. *The geometric median is invariant to the lever*: its cells move by at most +0.019, a breakdown-point argument appearing as a measurement. *The gate loosens under the lever*: the attacker's score barely moves (0.466 → 0.417) while honest false-rejection falls 0.799 → 0.476 – the honest population's own mean score slides through the fixed threshold (0.608 → 0.470), collapsing the discriminative gap (0.142 → 0.052). The gate loosens not because the attacker climbs toward the threshold but because the honest clients slide down through it; neither gated arm profits.

5.9. H4: the privacy-versus-admission operating curve

The operating curve is degenerate, and the extended sweep shows why (Figure 7). Across all twelve registered (σ, τ) points the honest false-rejection rate is pinned at 1.000 and the attacker admission reads 0.000 only because *nothing* is admitted – the gate rejects all ten clients in all thirty rounds, the model never leaves initialization, and the reported macro-F_1 is an untrained network's, below even the 0.121 majority floor; the ordered-trend test is correspondingly flat (Jonckheere–Terpstra $p = 0.500$), so H4 is not supported. The gate first opens between $\tau = 0.7$ and 0.8; past it the attacker is admitted 0.788–1.000 and macro-F_1 stays on the floor: no corner that was run buys the accuracy back (two caveats: the extension is pipeline+Krum only, and six runs diverged on non-finite scores – excluded, "div."). The argument is dimensional, not a tuning failure: on $\sim 10^5$-parameter updates, isotropic noise calibrated for local DP must overwhelm the update direction entirely – the scored cosine becomes noise – and even the weakest setting carries a near-vacuous budget (ε from 16.9 to 345). Update-level local DP and cosine admission are, on this architecture, mutually exclusive: a near-impossibility we quantify rather than a curve we tune.

5.10. H5: heterogeneity

Heterogeneity degrades the two weaker models and leaves the state-space model alone – not the pre-registered prediction (Table S16). From the near-uniform control to $\alpha = 0.1$, FedAvg CNN-LSTM macro-F_1 falls 0.683 → 0.499 and FedAvg spiking 0.190 → 0.118, while FedAvg Mamba does not move outside seed noise (0.770 → 0.798, in the wrong direction). Four of six ordered-trend tests survive Holm correction – both weaker models on both arms – while both Mamba trends stay flat at $n = 42$ runs per trend, so H5a fails its decision rule on architecture, not power. H5b fails too: exactly one of three cells clears the floor (spiking, $\delta = +0.88$), the CNN-LSTM cell inverts ($\delta = -0.51$) and Mamba's is flat. The conjunction fails on both legs for two different reasons; the state-space model is simply insensitive to label skew over the swept range – a finding, not a failure.

5.11. Consensus overhead

At the default committee a committing round costs exactly 180 messages and 167.8 ms of simulated wide-area latency against a training round orders of magnitude longer, so agreement is not the bottleneck (Table S17). The sweep's accuracy reading is no robustness story: pipeline+Krum is already collapsed at every committee size, and the moves between $M \in \{4, 7, 10\}$ are seed variance of a collapsed arm – overhead without accuracy return at this operating point. And the auditability claim is bounded exactly: pipeline arms append every round to a SHA-256 chain, but no `RunResult` carries an end-to-end verification, so auditability is a design claim with property-test coverage, not a measurement; the latency is a lower bound (Section 2.5).

5.12. Per-class behaviour

The macro average hides a rare-event problem (Figure 8). At the clean FedAvg reference the CNN-LSTM

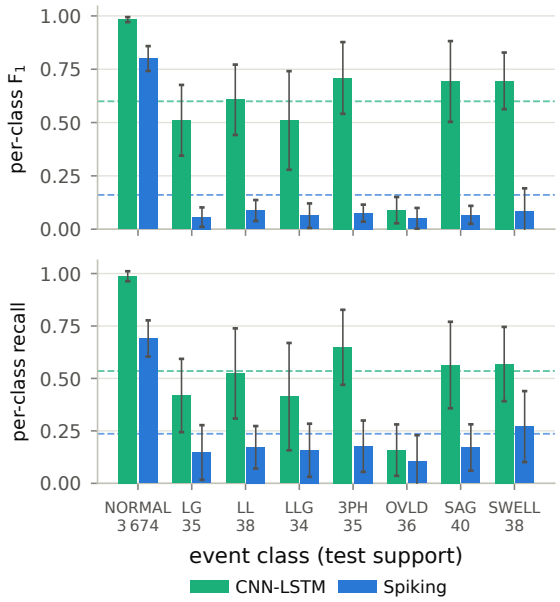


Figure 8: Per-class F_1 and recall (mean ± sd; dashed: macro score) with mean test support, clean FedAvg reference (T1a, random split, 15 seeds). Cells: Table S29.

carries NORMAL almost perfectly ($F_1 = 0.983$) but degrades on the fault and power-quality classes whose support is in the tens – down to $F_1 = 0.089$ on OVERLOAD – so its 0.600 is a rare-class average. The spiking model fails these outright: on FAULT_LG its recall of 0.147 comes with near-zero precision – the collapsed model sprays the rare label rather than never predicting it. These are statements about class rarity, not intrinsic detectability.

5.13. Ablations and the real-trace sanity check

Neither ablation rescues the spiking collapse or improves on FedAvg (Tables S19 and S20). Under delta coding, shortening the unroll helps – $T = 4$ gives the best clean macro-F_1 at 0.204 against $T = 8$'s 0.161 – so longer integration buys more SynOps for *less* accuracy, and rate coding pins at the trivial floor; the most favourable spiking configuration sits at about a quarter of Mamba's federated macro-F_1. FedProx is inert (largest paired gain +0.006; Mamba at $\mu = 0.1$ loses 0.150). On the MSU/ORNL real traces – a sanity check read narrowly: near-IID, effectively binary, random split, $n = 3$ – every arm runs to completion and the ordering *inverts* (spiking 0.996, Mamba 1.000, CNN-LSTM 0.703; pipeline 0.927–0.990 on the spiking and Mamba models, 0.486–0.726 on the CNN-LSTM): the pipeline is operable on real IEC 61850-adjacent traces, but the set does not validate the synthetic benchmark.

5.14. Statistical battery and hypothesis verdicts

Table S21 collects the complete battery and Table S22 the verdicts, and the scoreboard is unflattering but fully decidable: exactly two hypotheses are supported – H1b (the spiking energy ratio below one) and H3a (the defense-aware evasion) – while H1a, H2, H3b, H4, H5a and H5b are not, and the conjunctions fail with them; none is a power artefact any more. What the tables do support is an open, honestly scored benchmark carrying four durable findings: a three-corner model trade-off whose accuracy corner is *shared* between the two dense models; a negative result on elaborate Byzantine admission – robust averaging alone suffices and the gate buys auditability, not accuracy; an unpredicted heterogeneity result – the co-leading architecture is indifferent to label skew; and the one attack that breaks plain averaging – in-cone magnitude – against which the geometric median is invariant and the gate loosens.

6. Discussion

The interpretive frames in this section were fixed with the pre-registration, before any result existed, and are not adjusted to what the battery returned: where the result contradicts the registered expectation – and in three places it does – the frame is kept and the expectation is the thing that goes.

6.1. A three-way trade-off, not a single winner

The central picture is a trade-off, and the honest reading names the corner each architecture occupies – and on the headline chronological split the accuracy corner is *shared*, not held: the convolutional–recurrent reference and the selective state-space model tie at macro-F_1 0.596 and 0.590 ($p = 0.56$). The state-space model adds the highest centralized ceiling (0.724) at the largest compute (21.1 μJ arithmetic-only, p99 > 5× the co-resident envelope). The reference model is the *timing* choice: statistically indistinguishable accuracy, the only model whose p99 leaves any of that envelope (82%, GPU only). The spiking model is the *energy* choice: about 147× cheaper by SynOps accounting, but collapsed to 0.128 federated and 0.236 centralized – it does not learn this task at this budget. A deployment picks a corner; the benchmark's contribution is to make the corners measurable on a common substrate. Even that reading is complicated by the probe: the reference is not justified by *centralized* accuracy – only its federated arm edges the pooled probe.

6.2. Mechanisms: what happened, why, and what follows

Four findings earn their keep only if their mechanism is stated.

The random split inflates accuracy. What: every arm scored higher under the random permutation (the federated CNN-LSTM only within seed noise, −0.6%) – the spiking centralized ceiling by 60% – and the leakage-controlled split took it back. *Why:* neighbouring windows share 75% of their samples, so a high-capacity classifier on near-duplicate windows is doing lookup, not generalization. *Implication:* quote the chronological columns; the random split is kept as the diagnostic showing what an uncontrolled split credits to leakage – up to 0.19 macro-F_1 on the cells that had accuracy to inflate.

Physical heterogeneity looks Byzantine to a direction gate. What: at $\tau = 0.5$ the gate rejected 66–98% of *honest* updates while admitting the defense-aware attacker half the time. *Why:* the score is cosine distance to the round's median, and this federation is non-IID by construction – different feeders, load mixes and event populations push honest updates in legitimately different directions, and local-DP noise makes the same confusion literal. *Implication:* a direction-only statistic cannot separate the federation's own heterogeneity from malice; if a gate is deployed at all, calibration matters more than the statistic.

The admission gate degrades the robust aggregator by starving it. What: geometric-median aggregation alone holds its accuracy under the defense-aware attacker (0.609/0.726) while the same aggregator behind the gate drops to 0.420/0.051. *Why:* a robust average needs honest mass, a selector needs honest neighbours, and at $\tau = 0.5$ the gate discards most of the honest population – Krum's $n \geq 2f + 3$ condition is routinely unmet; opened to $\tau = 1.0$ the same gate costs almost nothing. *Implication:* a gate in front of a robust aggregator must be calibrated to admit; a gate that rejects most of the honest population is a denial-of-service on its own federation.

Accuracy, privacy, energy and latency pull in different directions. What: no arm takes every axis – the accuracy lead is shared between the only model that fits the timing envelope and the slowest model, while the cheapest per inference is the least accurate. *Why:* the desiderata bind different resources – capacity, sparse arithmetic, a compiled kernel, isotropic noise the direction gate cannot tell from an attack, and a jointly witnessed decision whose 180-message round buys a record, not accuracy. *Implication:* deployment chooses a corner, which is also why geometric-median-alone is the defensible default for the defense axis.

6.3. Energy against accuracy: what the ratio does not say

Three things must be read together. *First*, parity is a premise, not a bonus, and it failed – the paired Δmacro-F_1 interval $[-0.465, -0.417]$ sits far outside the ± 0.02 margin, so the energy comparison runs between models at different operating points; the literature had made parity look plausible [16], which is exactly why the negative result is worth reporting. *Second*, the ratio is arithmetic-only at 45 nm constants [3], excluding memory traffic – often the dominant term. *Third*, coding moves both axes at once: temporal-contrast emits events only where the signal changes – a strong prior on quasi-static telemetry – and is twelve times cheaper than rate coding *and* the only coding above the trivial floor, while thresholds below 0.1 fire on the instrument noise.

6.4. The defense asymmetry

The evasion half of the asymmetry is now decided rather than descriptive – admission rises from 0.000 to 0.53–0.87 at $p_{\text{Holm}} = 1.8 \times 10^{-4}$ in every cell – and the damage half splits on the aggregator: behind Krum the admitted attack costs measurably more than the rejected one on two of three models, while behind the geometric median it runs backwards, because a robust average is not moved by an update it was content to admit. The conclusion is narrower than the hypothesis and more useful: evading a cosine gate is easy and, against a robust average, close to pointless *at honest magnitude*.

The magnitude lever removes that qualification. What makes a single cosine threshold portable – scale-freeness – is exactly what makes it blind to norm: admitted magnitude translates directly into displacement, and plain averaging does not survive the lever ($0.598 \rightarrow 0.103$) while the geometric median barely registers it. The design recommendation therefore collapses to something simpler than the one we set out with: *deploy the geometric median on its own* – the only arm robust to the attack the gate cannot see – and if a cosine gate is kept for the auditable record, it needs a norm bound, which is a clipping rule, not an admission rule. A second, untested direction follows from the attacker's aiming error: an admission statistic not estimable from public information alone would widen that error and narrow the usable cone without changing the interface.

6.5. What the agreement layer buys, and what privacy costs

Byzantine agreement is not an accuracy mechanism here; what it remains is a *governance* mechanism – it makes the admission decision jointly witnessed rather than a server's private choice, and appends a hash-chained record, which in a federation of competing utilities is what a regulator or contractual counterparty asks for. The guarantee is bounded twice over: the chain is tamper-evident but does not stop a writer who re-hashes every successor, and this battery builds it in memory and discards it, so auditability is a design claim, not a measurement. The cost side rides in the same breath: the overhead figures are a *lower bound* (always-honest primary, no view change, no authentication).

On privacy the selection procedure has no solution on this architecture, because the curve is a point: at every $\tau \leq 0.7$ the gate rejects all ten clients in all thirty rounds; at $\tau \geq 0.8$ it admits the attacker essentially always and macro-F_1 stays at or below the trivial floor, at near-vacuous ε throughout – consistent with the theory [9] and an argument for the designs that route around the tension [25]: server-side or distributed noise, or admission on a quantity that survives isotropic perturbation.

6.6. Implications for a utility consortium

Read together, the frames suggest a deployment shape rather than a product. Federation is worth it: every federated arm beats its local-only floor under both protocols. Robust aggregation should be *selected* rather than composed – geometric-median-alone was the best-ranked defense arm and every layer on top cost accuracy; if an admission gate is deployed anyway, for auditability, it must be calibrated to admit. Agreement should be justified on governance, with its chain actually persisted and verified, which this battery did not do. Update-level privacy should not be composed with cosine admission on architectures of this size. And one recommendation reverses on the evidence: heterogeneity was expected to be the binding constraint, but the state-space model does not move across the swept skew – a consortium expecting severe label skew has an architectural answer before it has a federated-algorithm one. Whether that indifference survives beyond the swept range and beyond ten clients, we do not know; and the classifier should be chosen for the envelope of the hardware installed in the bay – an argument for evaluating event-driven models on event-driven hardware.

7. Limitations and conclusion

We list limitations at the granularity at which they could change a conclusion; each is a thing a reader should know before citing a number from this paper.

7.1. Limitations

Energy is an estimate, not a measurement. SynOps accounting prices counted synaptic events and analytic multiply–accumulate counts at published 45 nm constants [3] – arithmetic only, excluding memory access, data movement and peripherals, which on many accelerators dominate; the dense reference is analytic too, and no power was measured on any device.

The consensus overhead is a lower bound, and the ledger is not persisted. The primary is always honest, so no view change occurs; message authentication, batching, checkpointing and garbage collection are not priced; the wide-area latency distribution is fitted-synthetic [44]; and validators are simulated processes in one host. There is one in-memory chain, not one replica per validator, and no `RunResult` field persists it – chain integrity is covered by property tests, not by these runs.

Simulated data, not field data. Every synthetic window comes from a physics simulator; the fitted-synthetic rows are named in the realism audit (Tables S2–S5). The 60 s quasi-static cadence is the largest single simplification – sub-minute clearing sequences are unobservable, so episodes represent what one-minute RMS telemetry captures, not point-on-wave behaviour. No protection equipment is modelled: faults persist for their full drawn duration, so fault windows are more observable than in the field and fault-class accuracies are optimistic. Absolute macro-F_1 values outside the chronological-split tables are likewise optimistic ceilings – the sweeps, ablations and local-DP grid were not re-run under the controlled protocol. The f_{dev} channel is synthetic and event-coupled – a possible label shortcut, bounded by the probe ablation of Section 5.1 on a shallow model only. Fault location is bus-granular, photovoltaic modelling simplified, windows do not cross day boundaries, the class prior is a modelling choice, and one solver produced every window.

Scope of the battery. Seeds are fifteen at Tier 1 and the temporal controls, nine elsewhere, and the extension mixes two execution environments – bounded by a homogeneous-subset analysis and fixed by dated amendment. The client count is fixed at ten; the heterogeneity control is near-uniform, not uniform; hyperparameters were tuned once on clean FedAvg and frozen,

so the spiking collapse is a collapse *at this optimization budget* – a per-architecture search remains a live alternative explanation; no model was quantized; the temporal and magnitude batteries are post-registration controls; and root-dataset trust bootstrapping [8] is excluded by assumption, cited not re-run.

Threat model. Label-flip is dynamic per-batch noise, a weaker, higher-variance adversary than static poisoning. The adaptive attacker is defense-aware but not omniscient and never observes τ, so the H3b null does not separate an under-powered attacker from a working defense; the cone angle ς is never swept, so measured evasion is a lower bound; it wastes its first round by construction; adversary placement is structured and static; a single non-finite submission can deny the defense for a round – an open vulnerability we name rather than measure; Krum's $n \geq 2f + 3$ guarantee is void behind a tight gate; and the attacker blinds itself on a fail-closed round.

Reproducibility and per-class rarity. Bit-reproducibility is claimed on CPU only; per-run git provenance is mostly absent, so regeneration rests on manifests, configuration hashes (66.7% re-verifying canonically on the original battery) and seeded determinism (Section S4.6). The rarest fault class has single-digit support in some splits, so its recall is unstable and reported rather than smoothed – detecting it reliably needs a different sampling design.

7.2. Conclusion

GridSpike treats the four constraints of federated substation event classification – data that cannot be pooled, physical heterogeneity, an energy- and latency-bounded edge, and a federation with no trusted party – as one design problem, and stress-tests it end to end rather than claiming it robust. The programme was pre-registered and reported against its own decision rules, and several rules returned negative – which is the point of pre-registering them. On the leakage-controlled chronological split the two dense architectures tie at the top (macro-F_1 0.596/0.590); the spiking model, two orders of magnitude cheaper in estimated energy, is on the floor federated *and* centralized – it does not learn this task at this budget. The admission pipeline, ablated against the robust aggregator standing alone, buys an auditable, jointly witnessed decision rather than accuracy; local DP and cosine admission are incompatible at every usable privacy level; heterogeneity leaves the state-space model unmoved across the whole swept skew; the consensus layer's cost is a lower bound next to the accuracy it does not buy; and the auditability it does buy is a design property, not a measurement. Three directions follow: move the energy claim from analytic accounting to event-driven hardware; add a norm bound – or a statistic not estimable from public information – to any deployed cosine gate, since magnitude is the lever that breaks plain averaging while the geometric median absorbs it untouched; and release the generator's federations at SMART-DS scale [45], so federated grid learning can be compared on a common, physically grounded substrate.

Data availability

All data behind the paper's primary results are *generated* by the quasi-static OpenDSS simulation with synthetic event and measurement overlays described in Section 3, released as part of the software artifact together with the unmodified public test-feeder models it consumes [30] (BSD-3-Clause licence). One third-party dataset is used, and only for the real-trace sanity check of Section 5.13: the public MSU/ORNL power-system attack and disturbance dataset, used unmodified under its published terms; it supports no model or defense comparison in this paper. The federations behind every reported synthetic result are reproducible from source with a fixed seed:

```
python -m experiments.stage_data -out
data_battery
python -m experiments.run_all -dry-run
python -m experiments.gen_temporal_battery
python -m experiments.gen_magnitude_sweep
python -m experiments.manifest_tau
python -m experiments.gen_gap_x3
python -m experiments.run_all -data-dir
data_battery
  -out-dir runs -device cuda
python -m experiments.report -runs-dir
runs_snn_fixed runs
  runs_x3 runs_ssm_x3 runs_gap_x3
  -tables-dir paper/tables -summary
runs/summary.json
```

The first regenerates the labelled per-substation windows; the second enumerates every pre-registered run with its configuration hash; the next four re-emit the post-registration batteries as committed manifests (the chronological split, the magnitude, threshold and X3 gap sweeps; the remaining X3 manifests ship committed like the rest), all regenerating canonically; the last two execute the battery and rebuild the merged corpus, the tables and the statistical battery, deduplicating first by `run_id` and then on a semantic configuration key.

The generator, the generated federations, all per-run result records, the run manifest and the figure and table scripts are deposited in the public archive identified below, so that the tables and figures of Section 5 can be rebuilt without re-running the battery.

Code availability

The complete framework – simulation, models, federated engine, attack suite, consensus pipeline, experiment runner, analysis code and tests – is open source, together with the generated federations and every per-run result record, at `https://github.com/ufxa/gridspike` (code under MIT, data and results under CC BY 4.0); an archived release with a persistent identifier will be cited here on acceptance. The package keeps its development name (`fedgrid`) in the repository; the system is named GridSpike in this manuscript because that name is already in use for an unrelated published system.

Declaration of competing interest

[?? competing-interest statement: authors to confirm on the title page] The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. (The definitive statement, with author names, is on the separate title page.)

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the author(s) used **[?? name of tool / service]** in order to **[?? reason: e.g. draft and edit sections of the manuscript]**. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Note. Use of artificial intelligence *within the research methodology itself* is disclosed separately in Section 4.4, in accordance with the publisher's policy that methodological AI use belongs in the Methods rather than in this statement. No AI system is listed as an author, and no AI-generated image is presented as primary research data.

References

[1] H. B. McMahan, E. Moore, D. Ramage, S. Hampson, B. Agüera y Arcas, Communication-efficient learning of deep networks from decentralized data, in: Proceedings of the 20th International Conference on Artificial Intelligence and Statistics (AISTATS), Vol. 54 of PMLR, 2017, pp. 1273–1282, arXiv:1602.05629.

[2] J. K. Eshraghian, M. Ward, E. Neftci, X. Wang, G. Lenz, G. Dwivedi, M. Bennamoun, D. S. Jeong, W. D. Lu, Training spiking neural networks using lessons from deep learning, Proceedings of the IEEE 111 (9) (2023) 1016–1054, arXiv:2109.12894.

[3] M. Horowitz, 1.1 computing's energy problem (and what we can do about it), in: 2014 IEEE International Solid-State Circuits Conference Digest of Technical Papers (ISSCC), 2014, pp. 10–14. `doi:10.1109/ISSCC.2014.6757323`.

[4] T. S. Ustun, S. M. S. Hussain, A. Ulutas, A. Onen, M. M. Roomi, D. Mashima, Machine learning-based intrusion detection for achieving cybersecurity in smart grids using IEC 61850 GOOSE messages, Symmetry 13 (5) (2021) 826. `doi:10.3390/sym13050826`.

[5] J. Moore, Anomaly detection in IEC-61850 GOOSE networks: Evaluating unsupervised and temporal learning for real-time intrusion detection, arXiv preprint arXiv:2604.14233 (2026).

[6] M. Fang, X. Cao, J. Jia, N. Z. Gong, Local model poisoning attacks to byzantine-robust federated learning, in: Proceedings of the 29th USENIX Security Symposium, 2020, arXiv:1911.11815.

[7] V. Shejwalkar, A. Houmansadr, Manipulating the byzantine: Optimizing model poisoning attacks and defenses for federated learning, in: Proceedings of the 2021 Network and Distributed System Security Symposium (NDSS), 2021. `doi:10.14722/ndss.2021.24498`.

[8] X. Cao, M. Fang, J. Liu, N. Z. Gong, FLTrust: Byzantine-robust federated learning via trust bootstrapping, in: Proceedings of the 2021 Network and Distributed System Security Symposium (NDSS), 2021, arXiv:2012.13995.

[9] R. Guerraoui, N. Gupta, R. Pinot, S. Rouault, J. Stephan, Differential privacy and Byzantine resilience in SGD: Do they add up?, in: Proceedings of the 2021 ACM Symposium on Principles of Distributed Computing (PODC), 2021, pp. 391–401. `doi:10.1145/3465084.3467919`.

[10] T.-M. H. Hsu, H. Qi, M. Brown, Measuring the effects of non-identical data distribution for federated visual classification, arXiv preprint arXiv:1909.06335 (2019).

[11] M. A. Husnoo, A. Anwar, H. T. Reda, N. Hosseinzadeh, S. N. Islam, A. N. Mahmood, FeDiSa: A semi-asynchronous federated learning framework for power system fault and cyberattack discrimination, arXiv preprint arXiv:2303.16956 (2023).

[12] M. A. Husnoo, A. Anwar, H. T. Reda, N. Hosseinzadeh, S. N. Islam, A. N. Mahmood, FedDiSC: A computation-efficient federated learning framework for power systems disturbance and cyber attack discrimination, arXiv preprint arXiv:2304.03640 (2023).

[13] Y. Li, J. Liu, Z. Yang, G. Liao, C. Zhang, Clustered federated learning for generalizable FDIA detection in smart grids with heterogeneous data, arXiv preprint arXiv:2507.14999 (2025).

[14] Y. Venkatesha, Y. Kim, L. Tassiulas, P. Panda, Federated learning with spiking neural networks, IEEE Transactions on Signal Processing 69 (2021) 6183–6194. doi:10.1109/TSP.2021.3121632.

[15] M. V. Nguyen, L. Zhao, B. Deng, S. Wu, The robustness of spiking neural networks in federated learning with compression against non-omniscient byzantine attacks, arXiv preprint arXiv:2501.03306 (2025).

[16] K. Mahapatra, S. Lu, A. Sengupta, N. R. Chaudhuri, Power system disturbance classification with online event-driven neuromorphic computing, IEEE Transactions on Smart Grid 12 (3) (2021) 2343–2354. doi:10.1109/TSG.2020.3043782.

[17] M. Baruch, G. Baruch, Y. Goldberg, A little is enough: Circumventing defenses for distributed learning, in: Advances in Neural Information Processing Systems 32 (NeurIPS), 2019, arXiv:1902.06156.

[18] C. Xie, O. Koyejo, I. Gupta, Fall of empires: Breaking byzantine-tolerant SGD by inner product manipulation, in: Proceedings of the 35th Conference on Uncertainty in Artificial Intelligence (UAI), 2019, arXiv:1903.03936.

[19] M. García-Márquez, N. Rodríguez-Barroso, M. V. Luzón, F. Herrera, Krum federated chain (KFC): Using blockchain to defend against adversarial attacks in federated learning, arXiv preprint arXiv:2502.06917 (2025).

[20] Y. Li, C. Xia, C. Li, T. Wang, BRFL: A blockchain-based byzantine-robust federated learning model, Journal of Parallel and Distributed Computing 196 (2025) 104995. doi:10.1016/j.jpdc.2024.104995.

[21] I. Mironov, Rényi differential privacy, in: 2017 IEEE 30th Computer Security Foundations Symposium (CSF), 2017, pp. 263–275. doi:10.1109/CSF.2017.11.

[22] O. Aouedi, K. Piamrat, M. Südholt, HFedSNN: Efficient hierarchical federated learning using spiking neural networks, in: Proceedings of the International ACM Symposium on Mobility Management and Wireless Access (MobiWac), 2023, pp. 53–60. doi:10.1145/3616390.3618288.

[23] P. Blanchard, E. M. E. Mhamdi, R. Guerraoui, J. Stainer, Machine learning with adversaries: Byzantine tolerant gradient descent, in: Advances in Neural Information Processing Systems 30 (NeurIPS), 2017.

[24] K. Pillutla, S. M. Kakade, Z. Harchaoui, Robust aggregation for federated learning, IEEE Transactions on Signal Processing 70 (2022) 1142–1154. doi:10.1109/TSP.2022.3153135.

[25] X. Gu, M. Li, L. Xiong, DP-BREM: Differentially-private and byzantine-robust federated learning with client momentum, Proceedings on Privacy Enhancing TechnologiesArXiv:2306.12608 (2024).

[26] D. J. Beutel, T. Topal, A. Mathur, X. Qiu, J. Fernandez-Marques, Y. Gao, L. Sani, K. H. Li, T. Parcollet, P. P. B. de Gusmão, N. D. Lane, Flower: A friendly federated learning research framework, arXiv preprint arXiv:2007.14390 (2020).

[27] A. Gu, T. Dao, Mamba: Linear-time sequence modeling with selective state spaces, arXiv preprint arXiv:2312.00752 (2023).

[28] Y. Cui, M. Jia, T.-Y. Lin, Y. Song, S. Belongie, Class-balanced loss based on effective number of samples, in: 2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 9260–9269. doi:10.1109/CVPR.2019.00949.

[29] M. Castro, B. Liskov, Practical Byzantine fault tolerance, in: Proceedings of the Third Symposium on Operating Systems Design and Implementation (OSDI), USENIX Association, 1999.

[30] W. H. Kersting, Radial distribution test feeders, in: 2001 IEEE Power Engineering Society Winter Meeting, Vol. 2, 2001, pp. 908–912. doi:10.1109/PESW.2001.916993.

[31] I. Richardson, M. Thomson, D. Infield, C. Clifford, Domestic electricity use: A high-resolution energy demand model, Energy and Buildings 42 (10) (2010) 1878–1887. doi:10.1016/j.enbuild.2010.05.023.

[32] A. Hoke, R. Butler, J. Hambrick, B. Kroposki, Steady-state analysis of maximum photovoltaic penetration levels on typical distribution feeders, IEEE Transactions on Sustainable Energy 4 (2) (2013) 350–357. doi:10.1109/TSTE.2012.2225115.

[33] A. Woyte, R. Belmans, J. Nijs, Fluctuations in instantaneous clearness index: Analysis and statistics, Solar Energy 81 (2) (2007) 195–206. doi:10.1016/j.solener.2006.03.001.

[34] M. Lave, J. Kleissl, J. S. Stein, A wavelet-based variability model (WVM) for solar PV power plants, IEEE Transactions on Sustainable Energy 4 (2) (2013) 501–509. doi:10.1109/TSTE.2012.2205716.

[35] J. B. Dagenhart, The 40-ω ground-fault phenomenon, IEEE Transactions on Industry Applications 36 (1) (2000) 30–32. doi:10.1109/28.821792.

[36] V. D. Andrade, E. Sorrentino, Typical expected values of the fault resistance in power systems, in: 2010 IEEE/PES Transmission and Distribution Conference and Exposition: Latin America, 2010, pp. 602–609. doi:10.1109/TDC-LA.2010.5762944.

[37] IEEE Standards Association, IEEE Std 1159-2019: IEEE Recommended Practice for Monitoring Electric Power Quality (2019).

[38] M. H. J. Bollen, Understanding Power Quality Problems: Voltage Sags and Interruptions, IEEE Press, 2000.

[39] T. A. Short, Electric Power Distribution Handbook, CRC Press, 2004.

[40] International Electrotechnical Commission, IEC 61869-2: Instrument transformers – Part 2: Additional requirements for current transformers (2012).

[41] International Electrotechnical Commission, IEC 61869-3: Instrument transformers – Part 3: Additional requirements for inductive voltage transformers (2011).

[42] North American SynchroPhasor Initiative, Synchrophasor data quality, Tech. rep., NASPI Technical Report (2017).

[43] L. R. Gorjão, R. Jumar, H. Maass, V. Hagenmeyer, G. C. Yalcin, J. Kruse, M. Timme, C. Beck, D. Witthaut, B. Schäfer, Open database analysis of scaling and spatio-temporal properties of power grid frequencies, Nature Communications 11 (2020) 6362. doi:10.1038/s41467-020-19732-7.

[44] J. Aikat, J. Kaur, F. D. Smith, K. Jeffay, Variability in TCP round-trip times, in: Proceedings of the 3rd ACM SIGCOMM Conference on Internet Measurement (IMC), 2003. doi:10.1145/948205.948241.

[45] National Renewable Energy Laboratory, SMART-DS: Synthetic models for advanced, realistic testing – distribution systems and scenarios, Open Energy Data Initiative, submission 2981 (2021).

[46] T. Li, A. K. Sahu, M. Zaheer, M. Sanjabi, A. Talwalkar, V. Smith, Federated optimization in heterogeneous networks, in: Proceedings of Machine Learning and Systems (MLSys), Vol. 2, 2020, pp. 429–450, arXiv:1812.06127.

[47] J. Demšar, Statistical comparisons of classifiers over multiple data sets, Journal of Machine Learning Research 7 (2006) 1–30.
[48] S. Holm, A simple sequentially rejective multiple test procedure, Scandinavian Journal of Statistics 6 (2) (1979) 65–70.
[49] N. Cliff, Dominance statistics: Ordinal analyses to answer ordinal questions, Psychological Bulletin 114 (3) (1993) 494–509. doi:10.1037/0033-2909.114.3.494.